In [1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
# from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix,accuracy_score,precision_score,recall_
```

In [2]:
```
df = pd.read_csv('iris.csv')
```

In [3]:
```
df
```

Out[3]:

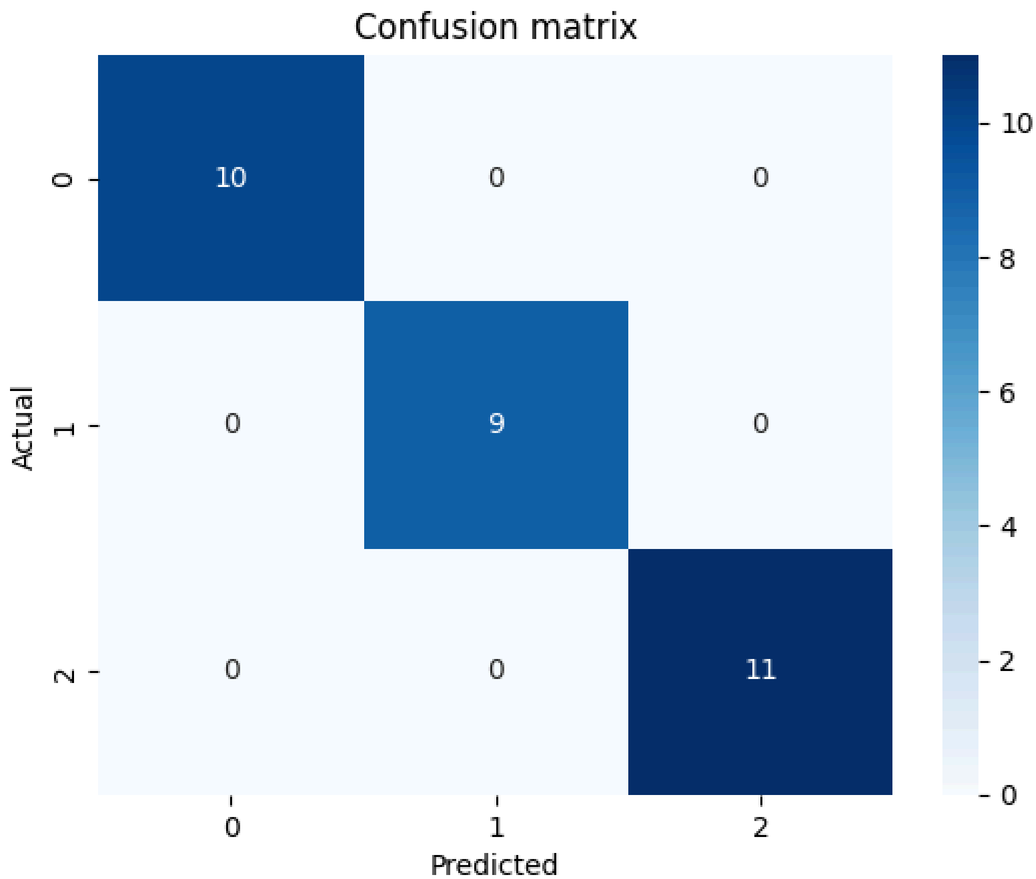
	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

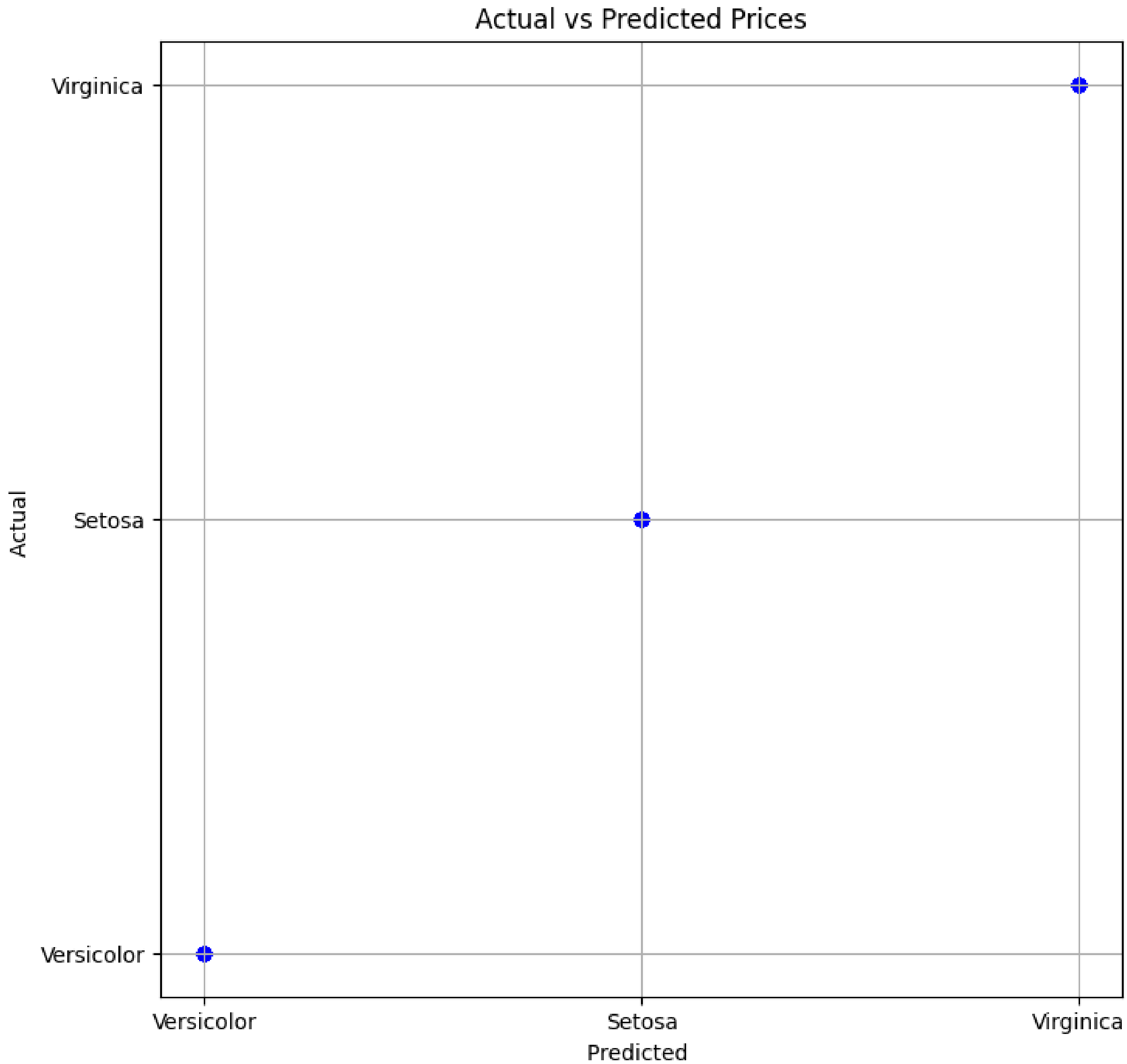
In [4]:
```
X = df.drop('variety', axis=1)
y = df['variety']

x_train,x_test,y_train,y_test = train_test_split(X,y,test_size = 0.2,random_state =
```

In [5]:
```
nb = GaussianNB()
nb.fit(x_train,y_train)
y_pred = nb.predict(x_test)
```

In [10]:
```
cm = confusion_matrix(y_test,y_pred)
sns.heatmap(cm,annot=True,cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion matrix")
plt.show()
```



In [18]:
```
plt.figure(figsize = (8,8))
plt.scatter(y_test,y_pred, color='blue')
plt.xlabel("Predicted ")
plt.ylabel("Actual ")
plt.title("Actual vs Predicted Prices")
plt.grid(True)
plt.show()
```



In [15]:
```
print(cm)
accuracy = accuracy_score(y_test,y_pred)
error = 1 - accuracy
precision = precision_score(y_test,y_pred,average='macro')
recall = recall_score(y_test,y_pred,average='macro')
print("Accuracy score: ",accuracy)
print("Error rate: ",error)
print("Precision score: ",precision)
print("Recall score: ",recall)
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
Accuracy score:  1.0
Error rate:  0.0
Precision score:  1.0
Recall score:  1.0
```

In [ ]:

In [ ]:

```

```